

Technical Challenge: Proposed solution

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[Repo link](#)

The problem

Given a raw documentation folder for multiple projects:

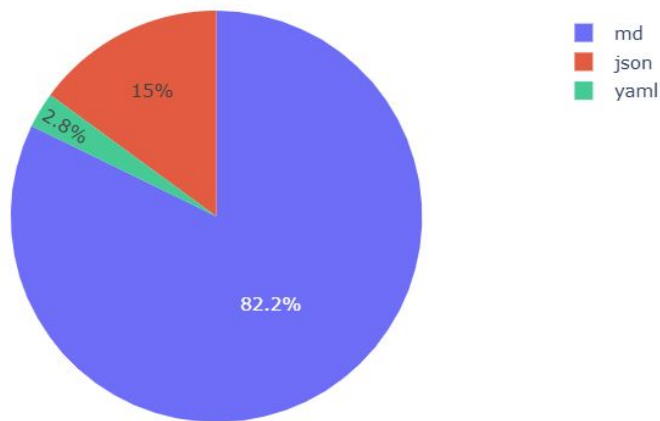
- Build a structured corpus of text.
- Build a structured Q&A dataset.

...For training LLMS using that data.

The data

Simple EDA to identify file types and how to create extraction codes for each one:

Distribution by data type in folder: /home/manuel/Job/MELI/docs_



- Total 107 files
 1. Focus on Markdown
 2. Focus on JSON
 3. Focus on yaml

*Note: be careful, since information is **not** uniformly distributed*

Corpus build

General structure

```
{
```

“Id”: simple_id,

“Name”: project_name,

“Path”: absolute_path,

“Version”: documentation/project version,

“Section”: subsection_of_documentation,

“Content”: selected_contentet

```
}
```

example



```
{ "Id": "2p-revenue-optimizer-api_0",  
  "Name": "2p-revenue-optimizer-api",  
  "Path":  
    "/home/manuel/Job/MELI/docs_raw/2p-revenue-optimiz  
er-api/0.2.2/guide/README.md",  
  "Version": "0.2.2",  
  "section": ["Spring Boot App model for Java 17"],  
  "content": "We provide a basic model for JDK 17 /  
Spring based web applications\\"}]
```

Markdown files

- Split by sections
- Encapsulate code snippets

```
1 # 2p-revenue-optimizer-api
2
3 # Spring Boot App model for Java 17
4
5 We provide a basic model for JDK 17 / Spring based web applications.
6
7 Please address any questions and comments to [VendorFlow Issue Tracker](https://github.com/vendorflow/vendorflow/issues).
8
9 ## Usage
```

SCOPE

```
12 The suffix of each VendorFlow **SCOPE** is used to know which properties file to use, it is identified from the last '-' of the name of the scope.
13
14 If you want to run the application from your development IDE, you need to configure the environment variable **SCOPE=local** in the app launcher.
15
16 The properties of **application.yml** are always loaded and at the same time they are complemented with **application-<SCOPE_SUFFIX>.yml** properties. If a property
17 is in both files, the one that is configured in **application-<SCOPE_SUFFIX>.yml** has preference over the property of **application.yml**.
18
19 For example, for the **SCOPE** 'items-loader-test' the **SCOPE_SUFFIX** would be 'test' and the loaded property files will be **application.yml** and
**application-test.yml**
```

Web Server

```
20
21
22 Each Spring Boot web application includes an embedded web server. For servlet stack applications, Its supports three web Servers:
23 * Tomcat (`spring-boot-starter-tomcat`)
24 * Jetty (`spring-boot-starter-jetty`)
25 * Undertow (`spring-boot-starter-undertow`)
26
27 This project is configured with Jetty, but to exchange WebServer, it is enough to configure the dependencies mentioned above in the build.gradle file.
```

- Section = ["Spring Boot App model for Java 17", "Usage", "SCOPE"]
- Content: escaped text from the red square.

JSON files

- Simple escaping, but preserving structure

```
{ "Name": "2p-revenue-optimizer-api",  
  "Path": "/home/manuel/Job/MELI/docs_raw/2p-revenue-optimizer-api/0.2.2/documentationTypes.json",  
  "Version": "0.2.2",  
  "section": "JSON",  
  "content":  
    "\\{\\\"guide\\\":\\\"https://vendorflow\\-docs\\.s3\\.amazonaws\\.com/2p\\-revenue\\-optimizer\\-api/0\\  
\\.2\\.2/guide/index\\.html\\\"\\}\\}"}
```

Yaml files

- JSON Normalization

```
1 openapi: 3.0.1
2 info:
3   title: fresh-products-api
4   description: Fresh Products API Documentation
5   contact:
6     name: Fresh Products Documentation
7     email: freshproducts@vendorsync.com
8   version: 0.0.1
9 servers:
10   - url: 'http://localhost:8080'
11     description: Scope local
12 paths:
13   /api/v1/fresh-products/inboundorder:
14     post:
15       tags:
16         - InboundOrder
17       summary: Insert batch in fulfillment warehouse
18       operationId: addBatchOfProductsToStock
19       requestBody:
20         content:
21           application/json:
22             schema:
23               $ref: '#/components/schemas/InsertBatchRequestDTO'
24             required: true
25       responses:
26         '201':
27           description: CREATED
28           content:
29             application/json:
30               schema:
31                 $ref: '#/components/schemas/InsertBatchResponseDTO'
```

example



- Section = ["Info", "title"]
- Content: fresh-products-api.

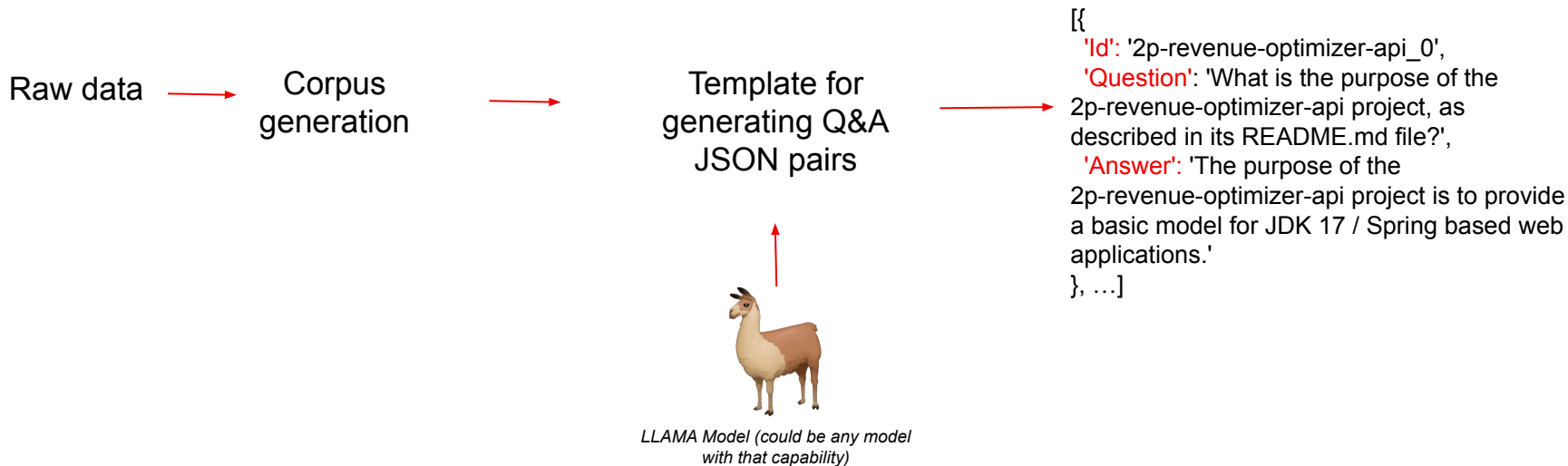
```
{ "Name": "zenith-keeper" ,
  "Path":
    "/home/manuel/Job/MELI/docs_raw/zenith-keeper/1.1.6/specs/swagger.yaml" ,
  "Version": "1.1.6",
  "section": ["info", "contact", "email"],
  "content": "freshproducts@vendorsync \\.com\\\"\\\" }" }
```


Q&A

Building the Q&A dataset

Once we have the corpus files, we can build the Q&A dataset:

- Automate process with LLAMA 3.1 (Possible licence issues)



Q&A Quality

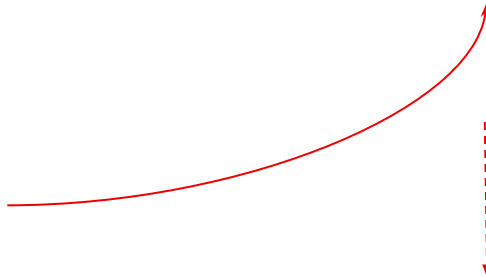
Llama 3.1 evaluation

Note: We are currently evaluating Llama 3.1 individually on the new [Open LLM Leaderboard 2](#) and will update this section later today. Below is an excerpt from the official evaluation from Meta.

Category	Benchmark	# Shots	Metric	Llama 3 8B	Llama 3.1 8B	Llama 3 70B	Llama 3.1 70B	Llama 3.1 405B
General	MMLU	5	macro_avg/acc_char	66.7	66.7	79.5	79.3	85.2
	MMLU PRO (CoT)	5	macro_avg/acc_char	36.2	37.1	55.0	53.8	61.6
	AGIEval English	3-5	average/acc_char	47.1	47.8	63.0	64.6	71.6
	CommonSenseQA	7	acc_char	72.6	75.0	83.8	84.1	85.8
	Winogrande	5	acc_char	-	60.5	-	83.3	86.7
	BIG-Bench Hard (CoT)	3	average/em	61.1	64.2	81.3	81.6	85.9
	ARC-Challenge	25	acc_char	79.4	79.7	93.1	92.9	96.1
Knowledge reasoning	TriviaQA-Wiki	5	em	78.5	77.6	89.7	89.8	91.8
	SQuAD	1	em	76.4	77.0	85.6	81.8	89.3
Reading comprehension	QuAC (F1)	1	f1	44.4	44.9	51.1	51.1	53.6
	BoolQ	0	acc_char	75.7	75.0	79.0	79.4	80.0
	DROP (F1)	3	f1	58.4	59.5	79.7	79.6	84.8

Taken from [LLAMA's HF page](#):

- In general terms, we can rely on big part of LLAMA's job, but it could fail.
- SQuAD metric: “““roughly””” 77% of “Accuracy” in Answering questions from a dataset similar to the one generated.
- Look at comprehension metrics and its ability to summarize



So, what's next?

Q&A Quality

LLAMA (or any other LLM) has done part of the hard job: generate the pairs.

Now assessment will be easy (easier).

Solution: Fact check ~ Reinforcement learning from Human Feedback (don't get confused, we don't want to "improve LLAMA")



```
{  
  'Id': '2p-revenue-optimizer-api_0',  
  'Question': 'What is the purpose of the  
2p-revenue-optimizer-api project, as described in its  
README.md file?',  
  'Answer': 'The purpose of the 2p-revenue-optimizer-api project  
is to provide a basic model for JDK 17 / Spring based web  
applications.'  
}, ...]
```

Idea:

1. Present the Q&A Pair with the original documentation to "Trained" audience with N people ($N > 1$, N must be an odd number, prime preferably).
2. Keep the "most" popular votation to know if a Q&A pair is indeed good, or LLAMA Failed

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}, ...]
```



If the pair is supported, we can use it for training



If the pair is NOT supported, we can discard it,
but maybe we can use it as *noise* for training.

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Corpus database

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  "Name": "2p-revenue-optimizer-api",
  "Path":
    "/home/manuel/Job/MELI/docs_raw/2p-revenue-optimizer-api/0.2.2/guide/README.
    md", "Version": "0.2.2",
  "section": ["Spring Boot App model for Java 17"],
  "content": "We provide a basic model for JDK 17 / Spring based web applications\\."
}
```

Join by ID to track
generation

Q&A database

```
{
  {
    'Id': '2p-revenue-optimizer-api_0',
    'Question': 'What is the purpose of the
    2p-revenue-optimizer-api project, as described in its
    README.md file?',
    'Answer': 'The purpose of the 2p-revenue-optimizer-api project
    is to provide a basic model for JDK 17 / Spring based web
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  }, ...
}
```

Opportunity of improvement: Change ID name from Q&A Database (set as foreign key) and add an unique Q&A ID (primary key).

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Task is clear: Generate (and keep) good data for any incoming process, this includes the traceability.

Q&A

(From your side)

Thank you